

Contents

TELECOM

60

Internet for all:
Ways to achieve

EDITOR : Ramesh Chopra

EDITORIAL
CORRESPONDENCE Phone: 011-26810601, E-mail: editor@efy.in
(Technical queries: eflab@efy.in)

SUBSCRIPTIONS &
MISSING ISSUES : Kin's Spares, New Delhi
Phone: 011-26376661, 2637662
E-mail: info@kisspares.com

BACK ISSUES,
BOOKS, CDs,
PCBs etc. : Kin's Spares, New Delhi
Phone: 011-26376661, 2637662
E-mail: info@kisspares.com

NEWS/STAND
DISTRIBUTION : Ph: 011-40596600
E-mail: efydc@efy.in

ADVERTISEMENTS
NEW DELHI
(HEAD OFFICE) : Ph: 011-26810601 or 02 or 03
E-mail: efynew@efy.in

MUMBAI : Ph: 022-24950047, 24928520
E-mail: efynew@efy.in

BENGALURU : Ph: 080-2560394, 2560023
E-mail: efybln@efy.in

PUNE : Ph: 08800295610, 09870682995
E-mail: efynew@efy.in

GUJARAT : Ph: 079-61344948
E-mail: efyahd@efy.in

JAPAN : Tandem Inc., Ph: 81-3-3541-4166
E-mail: japan@efy.in

SINGAPORE : Publicis Singapore Pte Ltd
Ph: +65-6836 2272
E-mail: singapore@efy.in

TAIWAN : J.K. Media, Ph: 886-2-87726780 ext. 10
E-mail: taiwan@efy.in

UNITED STATES : E & Tech Media
Ph: +1 860 536 6677
E-mail: usa@efy.in

Printed, published and owned by Ramesh Chopra, Printed at International Press o' Park Ltd, C-4 to C-11, Haryana Complex, Phase-II Extension, MODA-201 305, Gurgaon Road, Gurgaon, Haryana, on the first day of each month and published from 0-0171, Delhi India at New Delhi, New Delhi 110020. Copyright 2018. All rights reserved throughout the world. Reproduction of any material from this magazine in any manner without the written permission of the publisher is prohibited. Although every effort is made to ensure accuracy, no responsibility whatsoever is taken for any error in publishing errors. Articles that cannot be used are returned to the authors if accompanied by a self-addressed and sufficiently stamped envelope. But no responsibility is taken for any loss or delay in returning the material. Efy will be responsible for any wrong claims made by an advertiser. Disputes, if any, will be settled in a New Delhi court only.

SUBSCRIPTION RATES

Period	Newstand Price	You Pay	Digital	Overseas
Year	(₹)	(₹)	(₹)	Digital Print
Three	3600	2520	910	—
Two	2400	1800	648	US\$ 15
One	1200	960	345	US\$ 120

Please send payments only in favour of Efy Enterprises Pvt Ltd



INCLUDES 28-PAGE SPECIAL SUPPLEMENT:



To know more details
turn to Page No. 06

www.covestro.in

SMART WORLD

88 Does Your Office Intuitively Light Itself?

90 INTERVIEW: Reekrit Serai, Co Chair, CII: "The Whole (Education) Market is Virtually Untapped in India"

92 Solar Pumps: Renewable Energy For Better Agriculture

RESEARCH & DEVELOPMENT

18 SENSORS: Electronic Senses to Touch Us All

26 SOLAR: Next-generation Solar Power Technology (Part 1 of 2)

34 DESIGN: How to Design IoT Systems

38 EMBEDDED: Special Processors to Drive IoT and Wearables (Part 2 of 2)

45 INNOVATION: Touchless Drinking Water Dispenser Made in India

48 INTERVIEW: Paresh Patel, President & CEO, SLS: "Just Connecting Some Piece of Hardware to the Internet..."

76 TEST & MEASUREMENT: How Testing of IoT Systems Differs

GUIDE

52 BUYERS' GUIDE: Cellphone Signal Booster for Your Office

EVERYTHING YOU WANTED TO KNOW ABOUT...

55 SENSORS: Importance of Sensors in the Internet of Things

68 SECURITY: GeoFencing—Technology's Virtual 'Circle of Trust'

72 Internet of Things: Wireless Technologies for IoT

INDUSTRY

83 MARKET SURVEY: Industrial Automation and Control: Issues and Trends

96 PROJECT REPORT: Business Setup for e-Waste Recycling

DO-IT-YOURSELF



99 Gesture Controlled Wheelchair Using ATmega32

103 HVDC power supply design

106 Wireless VU meter

108 Smart switch box for mobile chargers and modems

110 Arduino-based front office visitor alarm

111 Multi-sensor data plotting using Arduino IDE

112 Analysis of transformer inrush current using Java

NEWS

12 Tech News

79 Industry News

86 New Products

EFY Plus DVD

EFY + DVD: Circuit Tool, Databases and Fun Stuff..... 82
CircuitMod: See What Really Happens in Your Circuit..... I
Logisim: A User-Friendly Logic Simulator VII
MariaDB: A Versatile Database Management System VII

Regulars

08 Feedback: Your suggestions

10 Q&A: Things you wanted to know